

Assignment 1: DIP

Unit I

- Q1. What is meant by Digital Image Processing? Explain how digital images are represented?
- Q2. What are the fundamental steps in Digital Image Processing?
- Q3. What are the components of an Image Processing System?
- Q4. Describe the elements of visual perception in DIP context.
- Q5. Explain image sampling and quantization process.
- Q6. Explain the basic relationships and distance measures between pixels in a digital image.

Unit II

- Q1. What is image enhancement? Discuss any four methods in pixel processing.
- Q2. Define histogram of a digital image. Explain how histogram is useful in image enhancement?
- Q3. Write about histogram specification/equalization.
- Q4. Write about Smoothing spatial filters.
- Q5. What is meant by the Gradient and the Laplacian? Discuss their role in image enhancement.
- Q6. Discuss the frequency domain techniques of image enhancement in detail.
- Q7. Explain Low Pass Filters in frequency domain.
- Q8. Explain High Pass Filters in frequency domain.
- Q9. Convert the following image into frequency domain using Fourier Transform by Kernel method, and use ideal high pass filter to smooth the image with $D_0=0.5$.
- $$f(x,y) = \begin{matrix} & 1 & 0 & 1 & 0 \\ & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & \\ & 1 & 0 & 1 & 0 \end{matrix}$$
- Q10. Explain how Laplacian is implemented in frequency domain.
- Q11. Write about high boost and high frequency filtering in frequency domain.